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Knowledge and practices about research integrity among Peruvian university students

Agueda Muñoz-del-Carpio-Toia^{1*}, Hardy Erick Guerra-Pérez², Lucía Begazo-Muñoz-del-Carpio³, Luis Fernando Ramos-Vargas⁴, Gonzalo Dávila-del-Carpio⁵, Yannina Melissa Peña-Pinto⁶, Patricio Díaz-del-Olmo-Calvo⁷, Enrique Mendoza-Carrera⁸ and Emanuele Valenti⁹

Abstract

Background Lately, the misconduct in research among scientist has been escalating. The use of artificial intelligence has been converted in a double ace card, on one side it has simplified the way of obtaining valious data for researchers in an easy way, but in the other hand, has promoted that researchers became careless about the scientific method. This situation has become a problem among university teachers since the students are tempt to obtain results in an easy way, leaving ethics aside. The objective of the study was to determine the knowledge and practices regarding scientific integrity among university students in Peru.

Methods A descriptive cross-sectional quantitative study was conducted, applying a validated survey on knowledge and practices regarding scientific integrity to 1,420 university students in the Southern Region of Peru, with 60.85% being women. The questionnaire consisted of 21 questions, and data were analysed using descriptive and inferential statistics. Robust statistics were applied, and comparisons were made based on gender, age, and region. Ages ranged from 17 to 29 years ($M = 21.14$, $SD = 2.72$).

Results The majority demonstrated high levels of knowledge and regular levels of practices regarding scientific integrity. Statistically significant differences were found based on gender (women > men), age (older than 23 years > younger than 22 years), and region. Inappropriate practices were identified in the use of artificial intelligence in academic work.

Conclusions It was concluded that university students have high levels of knowledge regarding scientific integrity; however, they exhibit regular and low levels of application in research projects.

Keywords Scientific integrity, Research integrity knowledge, Research integrity expertise, Undergraduate students

*Correspondence:

Agueda Muñoz-del-Carpio-Toia
amunozde@ucsm.edu.pe

¹Vicerrectorado de Investigación; Escuela de Postgrado, Universidad Católica de Santa María, Arequipa, Peru

²Escuela Profesional de Publicidad y Multimedia, Universidad Católica de Santa María, Arequipa, Peru

³Escuela Profesional de Publicidad y Multimedia; Vicerrectorado de Investigación, Universidad Católica de Santa María, Arequipa, Peru

⁴Escuela Profesional de Psicología, Universidad Católica de Santa María, Arequipa, Perú

⁵Facultad de Ciencias Farmacéuticas, Bioquímicas y Biotecnológicas, Universidad Católica de Santa María, Arequipa, Peru

⁶Departamento de Estudios Generales, Universidad Católica de Santa María, Arequipa, Peru

⁷Emory and Henry University, Virginia, USA

⁸Academia Nacional Mexicana de Bioética, Universidad Nacional Autónoma de México, Ciudad de México, México

⁹Division of Mental Health & Wellbeing, Warwick Medical School, University of Warwick, Coventry, UK



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Background

Universities are institutions entrusted with the task of shaping professionals with diverse skills essential for generating valid and replicable knowledge. This responsibility extends beyond training in methodological and ethical aspects of research; it requires constant vigilance to safeguard the scientific integrity of the entire university community. The recent 2024 version of the Declaration of Helsinki introduces for the first time the issue of scientific integrity as essential in research, with it being the responsibility of researchers as well as institutions to prevent scientific misconduct [1].

In recent years, researchers have faced increasing pressures, driven both by improvements in institutional rankings [2] and the academic motivation of researchers themselves. To be included in the National Scientific, Technological, and Technological Innovation Registry (RENACYT) of the National Council of Science, Technology, and Innovation of Peru (CONCYTEC), researchers must meet evaluation criteria, including the publication of articles in journals indexed in bibliographic databases such as Scopus and Web of Science [3].

Furthermore, universities have aligned their policies with the requirements of the National Superintendence of Higher Education (SUNEDU) to meet the quality standards for university research in the licensing process [4]. However, this normative adaptation by universities has not kept pace with the effective implementation and socialization of policies for preventing violations of scientific integrity across all universities [5].

Safeguarding scientific integrity in academia is a current challenge, as it is evident that tools are needed to address the prevention, identification, and management of scientific misconduct. This is especially crucial given the growing use of applications that incorporate generative artificial intelligence (AI) [6–8]. In addition to recent instances of inappropriate co-authorship in collaborative research due to article trafficking [9–11] and others related to scientific funding [12].

Currently, we observe an accumulation of evidence regarding scientific misconduct both globally [13–15] and at the national level [16–18]. The most recent modality in Peru involves the trafficking of authorships in scientific articles, as revealed in a journalistic report [19].

This recent discovery of authorship purchase has prompted CONCYTEC to express its disapproval and take immediate actions, such as publicly releasing a statement [20]. Additionally, CONCYTEC has initiated an update of essential documents in the national research system, including the National Code of Scientific Integrity [21], and an update to the RENACYT regulations as well as the publication of the regulation of sanctions for scientific misconduct [22].

In this scenario, it is essential to intensify efforts in academia to prevent such scientific misconduct. A suitable strategy involves generating evidence that describes the current situation. Therefore, the present study was proposed with the objective of determining the knowledge and practices regarding scientific integrity among university students in Peru.

Methods

Study setting and design

This study employed a descriptive, cross-sectional approach, utilizing a questionnaire administered to 1,420 university students from three universities in the Southern Region of Peru, specifically from faculties of health sciences and social sciences. 470 were university students from Arequipa, 470 studied in Cusco and 480 in Puno.

All participants interviewed were still university students. No graduates were interviewed. Authorization to use the instrument was not requested since the questionnaires were available on the web, but all of them have been correctly referenced.

Data collection tool and procedures

The questionnaire used was developed for this study (see supplementary data 1). A 21-item online questionnaire was employed, with information derived from four documents: (a) National Code of Scientific Integrity by CONCYTEC Peru [22]; (b) Questionnaire “Perceptions and Attitudes on Integrity and Misconduct in Research” [23]; (c) Questionnaire on Knowledge about Plagiarism [24]; and (d) “Questionnaire on Perceptions of Plagiarism by Biomedical Researchers: An Online Survey in Europe and China” [25].

The instrument offers response options ranging from Totally Disagree to Totally Agree. In terms of psychometric properties, the questionnaire’s content was assessed through a pilot study conducted by a panel of three experts in scientific integrity. They responded to an ad hoc questionnaire assessing the understanding and relevance of the items and determined the content validity. With their approval, a pilot test was applied.

To evaluate the psychometric properties, the internal structure of the original version of the instrument (30 items) was evaluated. First, an exploratory factor analysis with oblique rotation (oblimin) and the method of least residuals (minres) was applied considering three factors originally proposed: Knowledge, Attitudes and Practices. From the analysis performed, it was decided to remove nine items (8, 16, 17, 20, 21, 22, 23, 25 and 28) for: having low factor loadings (< 0.30), having factor loadings in two or more factors and that the content of the items did not fit the factor structure. A new exploratory factor analysis was applied with the 21 items (final version) considering two factors in the analysis. This two-factor model

explains 39% of the variance and the factor loadings were above 0.40. Two dimensions were identified: F1-Knowledge, referring to the evaluated individual's understanding of scientific integrity, comprised of the first nine items; and the second dimension named F2-Practices, addressing the recognition and avoidance of unethical research practices, formed by the remaining items. Supplementary Table 1 presents the factor loadings of the two-factor model analysis.

Additionally, a Confirmatory Factor Analysis was applied with the two-factor model using the WLSMV estimator. The results show an adequate fit ($\chi^2(188)=753.96$; CFI=0.91; TLI=0.89, RMSEA=0.10; SRMR=0.086). Although it should be noted that the RMSEA and SRMR fit indices are not in the adequate ranges, it is decided to maintain the two-factor model because of the congruence between the content of the items and their respective factors. In general, it can be affirmed that the model has an adequate fit. The factor loadings ranged from 0.29 to 0.89. Supplementary Fig. 1 shows the diagram of the 2-factor model.

For scoring, in the Knowledge dimension, responses of “Totally Agree” and “Agree” were considered correct (earning a score of 1 point), while the rest were marked as incorrect. In the Practices dimension, scores ranged from 1 (Totally Agree) to 5 (Totally Disagree), where a higher score indicated better ethical conduct in research practices.

The questionnaire was distributed through social networks among students from three universities in the Southern Region of Peru.

Data analysis

The data were digitized into a Microsoft Excel spreadsheet and subsequently imported into the statistical software R and its integrated development environment (IDE), R Studio. Prior to the analysis, the psychometric properties of the instrument were evaluated by assessing its content by a panel of three experts in scientific integrity. Additionally, a pilot test was applied to 300 participants, students from the three universities mentioned above, from the faculties of health sciences and social sciences. The internal structure was then evaluated by means of an exploratory factor analysis with oblique rotation (oblimin) and the method of least residuals (minres), finding two factors: F1 - Knowledge and F2 - Practices. With the two-factor structure, a confirmatory factor analysis was applied using the weighted least square mean and variance adjusted (WLSMV) estimator. Reliability was evaluated with the McDonald Omega coefficient. With the model obtained, the statistical analyses of the study were carried out.

Descriptive statistics were applied according to the variable type; frequency tables were presented for

Table 1 Description of participants

Variable	F	%
Sex		
Female	864	60.85
Male	556	39.15
Age		
Less than 18 years	282	19.86
19 to 22 years	688	48.45
Older than 22 years	450	31.69
Region		
Arequipa	470	33.1
Cusco	470	33.1
Puno	480	33.8

Table 2 Levels of knowledge and practices dimensions

Knowledge			Practices		
Levels	F	%	Levels	f	%
High	840	59.15	Good	434	30.56
Medium	311	21.90	Regular	568	40.00
Low	269	18.94	Poor	418	29.44

categorical variables, and for numeric variables, Mean, Standard Deviation, and results of normality tests were provided. For the analyses, statistical significance is established with a p-value < 0.05. During exploratory analysis, outliers were identified, leading to the decision to employ robust statistics using the WRS2 package. Winsorized correlations (20% trimming) were applied among the construct dimensions, followed by robust comparison tests (Yuen's test for two-group comparisons and F robust for comparisons involving three or more groups).

Results

The results are presented in the following order: firstly, participant characteristics; secondly, the levels of construct dimensions (knowledge and practices); thirdly, descriptive statistics for the dimensions along with the results of the normality test and the correlation between dimensions; fourthly, comparisons of dimensions based on sociodemographic variables (gender, age, and region); and finally, an item-by-item analysis according to the dimensions of the instrument.

A total of 1,420 university students participated, ranging in age from 15 to 29 years ($M=21.14$, $SD=2.72$) (see Table 1).

Table 2 presents the levels for the dimensions of knowledge and practices among the participants. It is observed that the predominant level of knowledge is high (59.15%), followed by medium and, finally, low. Regarding practices, the predominant level is regular (40.00%), followed by good (30.56%), and finally, poor (29.44%).

Table 3 presents the results of normality tests (Shapiro-Wilk). It is observed that the analyzed variables do not

Table 3 Description of construct dimensions

Dimension	M	Mdn	SD	SW	p
Knowledge	7.05	8	2.47	0.76	<0.001
Practices	41.74	42	7.96	0.98	<0.001

follow a normal distribution. The exploratory analysis also identified the presence of outlier cases.

As the studied variables do not follow normal distributions, robust statistics are applied. A correlation is conducted between the variables (rewinsorized = 0.22, $p < .001$). This indicates a statistically significant, positively correlated relationship between the variables with a low magnitude.

In Table 4, comparisons of dimensions (knowledge and practices) are presented based on sociodemographic variables (gender, age, and region). It was found that women showed a higher average in both knowledge and practices compared to men, with a larger effect size observed in the practices dimension. Statistically significant differences were also found in the knowledge dimension based on age. In post-hoc analyses, it was found that individuals older than 23 years, on average, scored higher compared to those younger than 18 years ($p < .05$) and the group aged 19 to 22 years ($p < .05$). No statistically significant differences were found in the practices dimension based on age. Finally, statistically significant differences were found in the knowledge dimension based on region. In post-hoc analyses, it was observed that the participant group from Puno scored higher on average than those from the Arequipa region ($p < .05$) and Cusco ($p < .05$). No statistically significant differences were found in the practices dimension.

In Table 5, the response distributions for items in the knowledge dimension are presented. It is observed that item 6, which refers to the deliberate alteration of study information, item 2, which pertains to the correct use of citation, and item 5, which involves “adding information about a procedure not performed,” were the items with the highest percentage of correct responses (> 85%).

Table 5 Distribution of responses for items in the knowledge dimension

Items		Incorrect	Correct
Item1	Does scientific misconduct refer to actions or omissions to good practices of scientific integrity?	18.66%	81.34%
Item2	Do you consider it an act of scientific misconduct to write texts using ideas from others without citing them?	15.00%	85.00%
Item3	Is it considered an act of scientific misconduct to use photos without citing the author (in case they have copyright)?	17.82%	82.18%
Item4	Do you consider the destruction of experiments to be an act of scientific misconduct?	22.46%	77.54%
Item5	Is indicating that a procedure was performed when it was not actually conducted considered an act of scientific misconduct?	15.00%	85.00%
Item6	Do you consider inaccurately presenting data, procedures, or research results to be an act of scientific misconduct?	14.58%	85.42%
Item7	Do you believe that the severity of scientific misconduct “depends on the intention to defraud, the degree of negligence, recurrence, and the severity of the impact of these behaviors”?	23.80%	76.20%
Item8	Do you consider granting co-authorship of an assignment or research work to someone who did not contribute significantly to be an act of scientific misconduct?	27.96%	72.04%
Item9	Is it an act of scientific misconduct to “not disclose conflicts of interest involving the institution where one works and the scientific research of others”?	39.37%	60.63%

Meanwhile, item 9, which addresses “reporting conflicts of interest,” item 8, which concerns “providing authorship

Table 4 Comparisons of construct dimensions by sociodemographic variables

	Knowledge			Practices		
	M	SD	S.E. (CI 95%)	M	SD	S.E. (CI 95%)
Sex						
Female	7.27	2.26	0.19 (0.07–0.29) **	42.5	7.36	0.26 (0.15–0.39) ***
Male	6.72	2.72		40.5	8.68	
Age						
Less than 18 years	7.00	2.34	0.12 (0.04–0.20) **	42.3	7.51	0.06 (0.01–0.14)
19 to 22 years	6.89	2.57		41.6	7.96	
Older than 23 years	7.34	2.37		41.6	8.25	
Region						
Arequipa	6.86	2.63	0.11 (0.05–0.17) **	41.3	8.03	0.07 (0.02–0.13)
Cusco	6.99	2.37		42.0	7.65	
Puno	7.3	2.39		42.0	8.20	

* $p < .05$; ** $p < .01$; *** $p < .001$

Table 6 Distribution of responses for items in the practices dimension

Items	TA	A	NAND	D	TD
Item 10 Would you include a professor's name as a co-author on your research poster if they haven't substantially contributed to the development of the scientific research?	6.41%	16.48%	16.76%	36.06%	24.30%
Item 11 Do you consider using generative artificial intelligence (ChatGPT or a similar tool) to draft texts for your assignments and/or research without disclosing its use is a lack of scientific integrity?	9.58%	21.55%	29.51%	22.32%	17.04%
Item 12 In a group project, do you expect your name to be added at the last minute as a supportive act by your peers to prevent you from failing?	3.59%	10.77%	21.41%	33.17%	31.06%
Item 13 In a group project, would you include the name of a friend or family member who did not participate in the work's development as a supportive act to prevent them from failing?	4.51%	12.75%	24.23%	32.39%	26.13%
Item 14 You don't have time to write a lengthy paper, and your best friend offers you their assignment. Would you make a copy with the assurance that you'll submit it to a professor who never reads students' papers?	4.01%	11.83%	23.38%	36.27%	24.51%
Item 15 Do you believe that the responsibility for the scientific integrity of a study lies solely with the mentor?	5.70%	20.00%	28.52%	31.27%	14.51%
Item 16 Should your teachers only consider plagiarism as the omission of citations for students conducting scientific research?	7.82%	24.08%	33.38%	23.38%	11.34%
Item 17 Do you think it is a normalized practice to use other people's ideas without citing them?	10.85%	30.42%	23.38%	21.62%	13.73%
Item 18 Do you use generative artificial intelligence (ChatGPT or a similar) to write your assignments or research papers?	6.34%	19.01%	28.94%	25.77%	19.93%
Item 19 Do you ask others to write an academic paper for you?	2.39%	9.23%	17.04%	36.41%	34.93%
Item 20 Your peers divide tasks and research; one of them contributes money to pay for typing the paper—do you consider them a co-author?	4.79%	14.01%	28.03%	32.04%	21.13%
Item 21 Do you use Photoshop to slim down and modify figures of individuals to make the research results look more appealing?	3.24%	10.77%	17.18%	31.90%	36.90%

TA Totally Agree, A Agree, NAND Neither Agree nor Disagree, D Disagree, TD Totally Disagree

without significant contribution,” and item 7, which focuses on the severity of scientific misconduct, had the highest percentage of incorrect responses.

In Table 6, the distribution of responses for the Practices dimension is presented. It is observed that items 17 and 16, which refer to citation in the text, and item 11, which relates to the “use of generative artificial intelligence like ChatGPT in the preparation of papers,” were the items associated with unethical practices, showing higher percentages of non-compliance with these scientific integrity practices.

Discussion

It was found that the majority of respondents (6 out of 10) exhibited high levels of knowledge regarding scientific integrity, but only 40% (4 out of 10) demonstrated regular levels of ethical research conduct practices. This inconsistency between knowledge and practices highlights that the participating students, at a cognitive level, understand which behaviours are correct or incorrect but do not equally implement them. In other words, they recognize unethical practices but still engage in them. Considering that universities are the primary producers of scientific knowledge, used to inspire new research and inform decision-making addressing the problems and needs of the population [26], it would be expected that all this information constitutes evidence obtained through rigorous methods and in line with ethical and

deontological principles that characterize research. However, concerns persist regarding the frequency of questionable behaviours in science, emphasizing the need to initiate efforts in student education to “strengthen supervision and processes of scientific integrity” [27].

Regarding comparisons based on gender, it was found that women exhibit higher scores in both the knowledge and practices dimensions compared to men. This could imply that women, in general, are knowledgeable about and actively engage in ethical behaviours in research activities. This aligns with other studies [30], who observed that women tended to adhere better to ethical aspects when conducting research compared to men. Although the differences (effect sizes) are small, the result is consistent with research on how morality operates in relation to gender.

It was found that individuals aged 22 years and older had, on average, higher scores than those under 21 years. Considering that the participant sample consisted of university students, and that students aged 23 or older are closer to conducting their research work for obtaining their professional degree, it is understandable that they may be more aware of the importance of scientific integrity compared to students who do not yet have such a close experience related to scientific research. Interestingly, it is observed that the group aged 19 to 22 years had, on average, lower scores compared to the other two groups. This particular group is already studying in the

middle years of their professional career, and the pressure from academic demands, along with a permissive environment, may contribute to students having lax knowledge about scientific integrity.

Another interesting result is that in the dimension of knowledge and practices according to age and region, statistically significant differences were only found in the first dimension. This result is noteworthy, as mentioned, one group has greater knowledge than the other two; however, all groups have the same scores in practices. This could mean that university students, despite having knowledge about good scientific conduct, do not put it into practice, as evidenced by all groups having the same average score.

Overall, the study participants showed high levels of knowledge of scientific integrity but had greater difficulty recognizing “conflicts of interest” as well as “co-authorship issues and the severity of scientific misconduct,” which are also recognized as violations of scientific integrity [28]. In terms of the practices dimension, it was found that participants “do not adequately report bibliographic citations” and use artificial intelligence in the elaboration of tasks and research papers inappropriately. University students “agree” and are “totally in agreement” in using generative AI applications that include “Large Language Models (LLMs)” like ChatGPT to write tasks or research papers. In this regard, the writing of texts and research manuscripts raises various issues related to the scientific integrity of authorship, which also need to be addressed in the university setting. The inappropriate use of generative artificial intelligence in scientific paper writing could expose students to plagiarism [29, 30].

Regarding practices, it is noteworthy that more than 41% of students consider it a normalized practice to use ideas from others without citing them. This finding aligns with other research related to plagiarism in university students. Our results coincide with those obtained by Pérez and his team, who surveyed 340 university students in Arequipa, Peru, and found that plagiarism was allowed, tolerated, and considered normal behaviour at the university [31].

Now, in our results, it was found that participants demonstrated greater knowledge of scientific integrity, contrasting with practices towards good scientific conduct. A significant number of participants do not recognize and are not avoiding unethical practices in their research. These results contrast with those reported in a study conducted with 388 students from a university in Croatia, where it was found that students were not very familiar with theoretical aspects of plagiarism; however, they showed positive attitudes toward avoiding plagiarism [32].

Regarding the current perspectives and strategies for universities to address the results, we consider that

universities must make significant changes to promote a culture that recognizes good practices and discourages research misconduct [33] through transversal programs of teaching scientific integrity with theoretical and practical aspects such as the review of emblematic cases, simulation of cases, etc. In this regard, a study conducted with 613 participants from faculties of the 100 best research universities in the United States on their perception of the effectiveness of intervention strategies on scientific misconduct, found that those interviewed prioritized punitive sanctions against researchers who commit violations of scientific integrity, followed by prevention efforts, training, and comprehensive approaches [33].

Universities must implement institutional policies that prevent and control scientific misconduct, assuming that any lack of scientific integrity can “undermine decades of research” [34] losing public confidence in the objectivity of the results and credibility of the university itself, even more so if institutions only care about minimizing the damage to the reputation of the institution and do not make effective changes to prevent and manage the problem.

Finally, another noteworthy finding is that participants, regarding potential authorship issues, expressed their rejection of gift authorship. Specifically, 60.36% disagreed with including a professor’s name on a poster as a co-author of research if they did not substantially contribute to its development. Likewise, 64.23% predominantly disagreed with suggesting that they, or their friends and family (58.52%), be included in tasks or work they did not participate in to avoid being disapproved. These results are encouraging, as university students consider authorship recommendations based on their academic contribution, as indicated by most authorship guidelines [35–37].

The main contribution of the study is to show that it is not enough to ensure theoretical knowledge on issues related to scientific integrity; It is also necessary that universities must worry about implementing permanent spaces for practical recognition and consultation of situations that could affect scientific integrity.

Study limitations

The study is subject to limitations, particularly regarding the studied sample, as it exclusively represents university students in the Southern region of Peru. Therefore, the findings may not be generalized to other student populations, especially those pursuing careers unrelated to health and social sciences. Furthermore, we recognize that currently in Peru scientific integrity is considered a sensitive and delicate issue. University students have greater access to information about scientific integrity, due to the spread of television reports of cases

of factories buying and selling scientific articles, as well as recent changes in integrity policies, such as the new “Code of Scientific Integrity and Regulation of sanctions for scientific misconduct”. In this context, it is likely that the participants are biased when answering the questionnaire given this greater access to information, since they have greater knowledge than university students from other countries would normally have.

Conclusions

It was concluded that university students have high levels of knowledge regarding scientific integrity; however, they exhibit regular and low levels of application in research projects.

Abbreviations

RENACYT	National Scientific, Technological, and Technological Innovation Registry
CONCYTEC	National Council of Science, Technology, and Innovation
SUNEDU	National Superintendence of Higher Education
AI	Artificial Intelligence

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12910-025-01344-9>.

Supplementary Material 1.

Supplementary Material 2.

Supplementary Material 3.

Supplementary Material 4.

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Authors' contributions

AMCT and EV conceptualized and designed the study. AMCT, HEGP, LFRV, LBMC, YMPP, PDOC were involved in data acquisition, formal analysis, and data interpretation. AMCT, EV, LFRV, LBMC, PDOC, EMC, GDdC contributed to the drafting of the original manuscript, while YMPP, HEGP provided input during the revision of the original draft. All authors participated in the review and approval of the final manuscript.

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Data availability

The data that support the findings of this study are available just for the authors, since restrictions are applied to the availability of these data, because the informed consent state that they were not publicly available. However, upon reasonable request and with permission from the University Research Ethics Committee it could be accessed for your review.

Declarations

Ethics approval and consent to participate

Informed consent was obtained from all university students who participated in the study, there were no minors under 17 years old (see supplementary data 2). The participation of university students was voluntary and anonymous. The questionnaire did not include sensitive personal data. Prior to completing the questionnaire, participants were presented with informed consent. The project received a favorable review with approval number

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Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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